Informatica® Cloud Application Integration
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Google BigQuery V2 Connector Guide

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Preface

Read the *Google BigQuery V2 Connector Guide* to learn how to set up and use Google BigQuery V2 connections. Use a *Google BigQuery V2 connection* to read from or write to Google BigQuery V2 by using Application Integration. This guide assumes that you have an understanding of Google BigQuery V2 concepts.

CHAPTER 1

Introduction to Google BigQuery V2 Connector

You can use Google BigQuery V2 Connector to read data from and write data to Google BigQuery. It can handle both structured and unstructured data, and provides real-time streaming ingestion for continuous analytics.

You can create a Google BigQuery V2 connection and use it in a process. When you use this connector to create and run a process, the Secure Agent reads from and writes data to Google BigQuery based on the process and Google BigQuery connection configuration. You can move data from any data source to Google BigQuery.

Google BigQuery overview

BigQuery is a fully managed, AI-ready data platform that helps you manage and analyze your data with built-in features such as, machine learning, search, geospatial analysis, and business intelligence. You work seamlessly with both structured and unstructured data, supporting open table formats like Apache Iceberg, Delta, and Apache Hudi. BigQuery streaming enables continuous data ingestion and real-time analysis, while its scalable engine lets you query terabytes in seconds and petabytes in minutes. You discover, curate, and manage data quality using built-in governance tools powered by Dataplex Universal Catalog. You also share data and AI assets securely across your organization with access control. BigQuery's architecture separates storage and compute layers, which communicate efficiently over Google's high-speed network to deliver fast, flexible analytics.

CHAPTER 2

Google BigQuery V2 Connections

Create a Google BigQuery V2 connection to read data from a Google BigQuery source and write data to a Google BigQuery target. When you create a Google BigQuery V2 connection, you can configure a connection mode based on how you want to read and write the data.

After you create a Google BigQuery V2 connection, you can validate, test, and save the connection.

You can then publish the Google BigQuery V2 connection and click the **Metadata** tab to view the generated process objects for the connection.

Before you begin

Before you configure a connection, ensure that you download the Google service account key file in JSON format. The service account key file is created when you create a Google service account.

You require the client email, private key, and project ID from the service account key JSON file to create a Google BigQuery connection.

Basic connection properties

The following table describes the basic properties that you can configure on the **Properties** tab of the connection creation page:

Property	Description
Name	Required. Unique name for the Google BigQuery V2 connection that identifies it in the Process Designer. The name must start with an alphabet and can contain only alphabets, numbers, and hyphens (-).
Location	Required. The location of the project or folder where you want to save the connection. Click **Browse** to select a location. If the **Explore** page is currently active and a project or folder is selected, the default location for the connection is the selected project or folder. Otherwise, the default location is the location of the most recently saved asset.
Description	A description of the connection.

Property	Description
Type	Required. The type of connection you want to use for the connector or service connector. Select **Google BigQuery V2**.
Runtime Environment	Required. The runtime environment for the connection. Select the Secure Agent group or the Secure Agent machine where the connection must run.
Connection Test	Displays the result of the last connection test.
OData-Enabled	Specifies whether OData feeds are enabled for the connection. Select **Yes** to enable OData feeds. If you select **Yes**, you must specify either the allowed users or the allowed groups that can access the connection at design time. Default is **No**.
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
OData Cloud Access Enabled	Specifies whether you can access data from the OData service by using the Cloud endpoint URLs or the Secure Agent endpoint URLs. If you enable OData and configure the connection to run on a Secure Agent machine or a Secure Agent group, you can choose to disable access to the Cloud endpoint URLs for security purposes. Select **No** to disable access to the Cloud endpoint URLs. If you select **No**, you cannot access data from the OData service by using the Cloud endpoint URLs. You can access data only by using the Secure Agent endpoint URLs. Select **Yes** to access data from the OData service by using both the Cloud endpoint URLs and the Secure Agent endpoint URLs. Default is **Yes**. **Note:** If you set the **OData-Enabled** option to **No**, the value that you set for the **OData Cloud Access Enabled** option does not apply because Application Integration does not generate the OData endpoint URLs. For more information, see Endpoint URLs for OData-Enabled Connections.
Allowed Users for OData	The users that have access to the connection at design time. You can specify more than one user. After you specify the first value, press the Enter key or the Comma key, and then specify the next value.
Allowed Groups for OData	The user groups that have access to the connection at design time. You can specify more than one user group. After you specify the first value, press the Enter key or the Comma key, and then specify the next value.

Property	Description
Enable CORS for Domains	The additional allowed domains from which API consumers can invoke the associated APIs. API consumers running a web application running at one origin can access selected resources from origins that you mention in the **Enable CORS for Domains** field. Enter the wildcard character (*) to enable CORS for all domains. Enter a comma-separated list of domains to enable CORS for specific domains. For example, enter the following phrase to enable CORS for the abc.domain.com and xyz.domain.com domains: `abc.domain.com, xyz.domain.com`
Enable CORS for Verbs	The allowed verbs that API consumers can use to invoke the APIs from the domains that you specified in the **Enable CORS for Domains** field. Enter a comma-separated list of verbs to enable CORS for specific verbs. For example, enter the following phrase to enable CORS for the POST, PUT, and GET verbs: `POST, PUT, GET` Default is all verbs.

Authentication

You can configure the Service Account authentication method to access Google BigQuery. Select the **Service Account** authentication method and configure the authentication-specific parameters.

Service Account authentication requires, at a minimum, your Google BigQuery service account email, service account key, and project ID.

The following table describes the basic connection properties for Service Account authentication:

Property	Description
Service Account Email	Required. The client_email value from the Google service account key JSON file.
Service Account Key	Required. The private_key value from the Google service account key JSON file. To use the private key from the JSON file, enter the entire block of text found within the double quotation marks in the `private_key` field, starting with `-----BEGIN PRIVATE KEY-----\n` and ending at `-----END PRIVATE KEY-----\n`.
Project ID	Required. The project_id value from the Google service account key JSON file. If you have created multiple projects with the same service account, enter the ID of the project that contains the dataset that you want to connect to.

Note: If you want to validate the credentials for the **Service Account Email**, **Service Account Key**, and **Project ID** fields when you test the connection, set the flag `CredentialValidation:true` in the **Provide Optional Properties** field under advanced settings.

Google BigQuery V2 connection properties

When you create a Google BigQuery V2 connection, you must configure the connection properties.

The following table describes the Google BigQuery V2 connection properties:

Property	Description
Advanced Settings	
Enable BigQuery Storage API	Uses the Google BigQuery Storage API to stage the files when you read or write data. Default is **No**.
Storage Path	Path in Google Cloud Storage where the Secure Agent creates a local stage file to store the data temporarily. The Secure Agent uses this storage when it reads data in staging mode or writes data in bulk mode. Use one of the following formats: - `gs://<bucket_name>` - `gs://<bucket_name>/<folder_name>` When you enable cross-region replication in Google BigQuery, enter a Google Cloud Storage path that supports dual region storage. This property is not applicable if you use Google BigQuery Storage to stage the files.
Connection mode	The mode that you want to use to read data from or write data to Google BigQuery. Select one of the following connection modes: - **Simple**. Flattens each field within the Record data type field as a separate field in the process. - **Hybrid**. Displays all the top-level fields in the Google BigQuery table including Record data type fields. Google BigQuery Connector displays the top-level Record data type field as a single field of the String data type in the process. Default is **Simple**. This property is applicable if you use Google Cloud Storage to stage the files.
Use Legacy SQL for Custom Query	Select this option to use legacy SQL to define a custom query. If you clear this option, use standard SQL to define a custom query. This property is applicable if you use Google Cloud Storage to stage the files. This property doesn't apply if you configure the Google BigQuery V2 connection in hybrid mode. Default is **No**.
Dataset Name for Custom Query	The name of the Google BigQuery dataset that you want to specify in a custom query.
Schema Definition File Path	Directory on the Secure Agent machine where the Secure Agent creates a JSON file with the sample schema of the Google BigQuery table. The JSON file name is the same as the Google BigQuery table name. Alternatively, you can specify a storage path in Google Cloud Storage where the Secure Agent creates a JSON file with the sample schema of the Google BigQuery table. You can download the JSON file from the specified storage path in Google Cloud Storage to a local machine. When you use a serverless runtime environment, specify a storage path in Google Cloud Storage. This property is applicable if you use Google Cloud Storage to stage the files.
Region ID	The region name where the Google BigQuery dataset that you want to access resides. **Note:** Ensure that you specify a bucket name or the bucket name and folder name in the **Storage Path** property that resides in the specified region.

Property	Description
Staging Dataset	The Google BigQuery dataset name where you want to create the staging table to stage the data. You can define a Google BigQuery dataset that is different from the source or target dataset. This property is applicable if you use Google Cloud Storage to stage the files.
Provide Optional Properties	Comma-separated key-value pairs of custom properties in the Google BigQuery V2 connection to configure certain source and target functionalities.
Retry Strategy	
Enable Retry	Select this option if you want the Secure Agent to attempt a retry to receive the response from the Google BigQuery endpoint. You can configure the retry strategy to read data from Google BigQuery in direct or staging mode and write data to Google BigQuery in bulk mode.
Maximum Retry Attempts	The maximum number of retry attempts that the Secure Agent performs to receive the response from the Google BigQuery endpoint. If the Secure Agent fails to connect to Google BigQuery within the maximum retry attempts, the connection fails. Default is **6** attempts.
Initial Retry Delay	The initial wait time in seconds before the Secure Agent attempts to retry the connection. Default is **1** second.
Retry Delay Multiplier	The multiplier that the Secure Agent uses to exponentially increase the wait time between successive retry attempts up to the maximum retry delay time. Default multiplier is **2**. You can also use fractional values.
Maximum Retry Delay	The maximum wait time in seconds that the Secure Agent waits between successive retry attempts. Default is **32** seconds.
Total Timeout	The total time duration in seconds that the Secure Agent attempts to retry the connection after which the connection fails. Default is **50** seconds.
Object Properties	
Dataset	Required. Name of the dataset that contains the source table and target table that you want to connect to. **Note:** Google BigQuery supports the datasets that reside only in the US region.
Object Filter	Configure an object filter to add conditions to filter the objects. You can specify multiple object names separated by a comma. Application Integration populates the metadata only for the specified objects and their related objects.

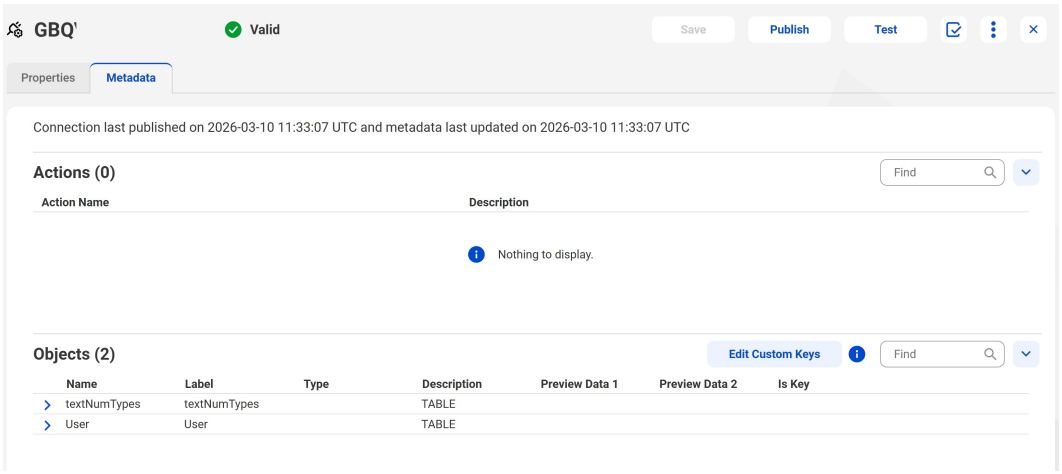
Note: Ensure that you specify valid credentials in the connection properties. A connection test is successful even if you specify incorrect credentials in the connection properties.

Google BigQuery V2 connection metadata

After you create a Google BigQuery V2 connection, you can validate and save the connection. You can then publish the Google BigQuery V2 connection and click the **Metadata** tab to view the generated process objects for the connection.

When you publish a Google BigQuery V2 connection, a list of **Objects** appear in the **Metadata** tab. The **Objects** section displays the entities along with the column metadata for the schema you specified in the connection. You can also preview the data.

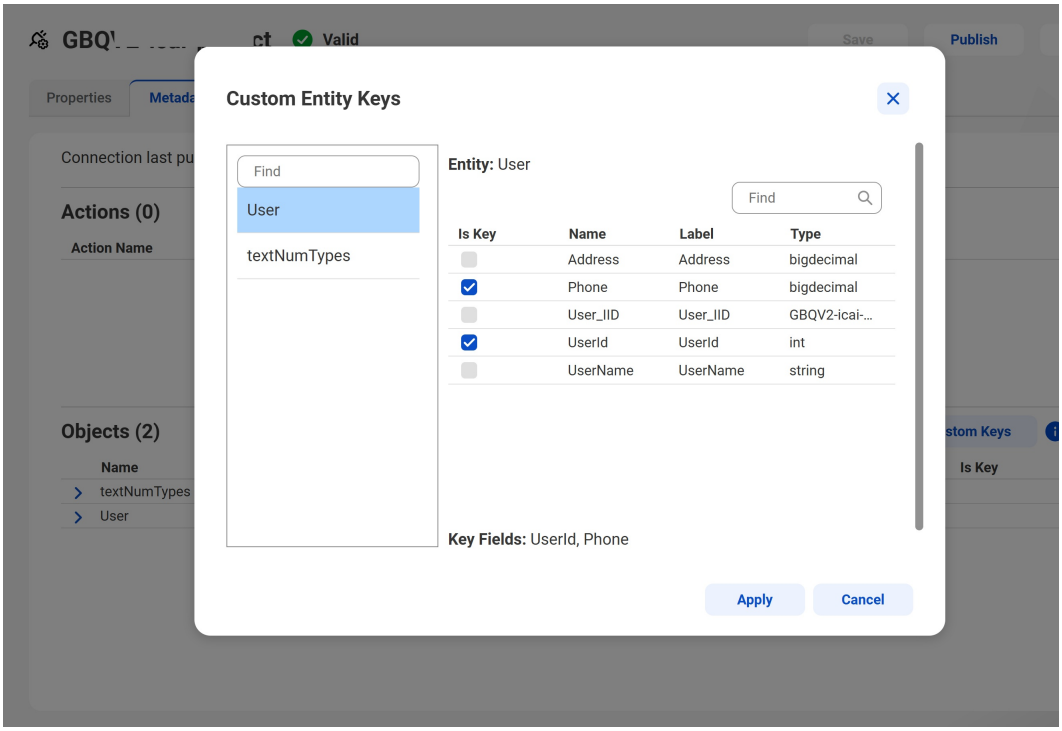
The following image shows the **Metadata** tab for a published Google BigQuery V2 connection:



You use custom keys to define primary keys for a table. As Google BigQuery V2 doesn't enforce primary key constraints, you provide primary keys as custom keys.

Note: You need primary keys to perform update and delete operations on a table.

To edit the custom key, click **Edit Custom Keys**. The **Custom Entity Keys** page appears as shown in the following image:



Select the required custom keys for each entity and republish the connection to apply the changes.

When you publish a Google BigQuery V2 connection, the **Metadata** tab is refreshed by default. You can click **Publish (Skip Metadata Refresh)** to skip the metadata refresh and reduce publishing time.

Using a Google BigQuery V2 connection in a process

You can perform create, read, update, and delete (CRUD) operations on Google BigQuery tables by using a Google BigQuery V2 connection in your process. Use these operations to manage data in Google BigQuery tables directly from Application Integration workflows.

Guidelines

Consider the following guidelines when you work with Google BigQuery V2 connections in a process:

- Ensure that you have the appropriate permissions to perform operations on the Google BigQuery tables.
- Use process fields to pass dynamic values to queries and operations.

Read data from a Google BigQuery table

You can use a SQL WHERE clause within an Assignment step to read data from a Google BigQuery table. You can filter the data, select specific fields, and assign the query results to process output fields that you use in subsequent steps. After you create a Google BigQuery V2 connection, you can validate, test, and save the connection. You can then publish the Google BigQuery V2 connection and click the **Metadata** tab to view the generated process objects for the connection.

Perform the following sample steps to read data from a Google BigQuery table:

1. In the Process Designer, open the process.

2. Create a process output field to store the query results.
   For example, create a **User** object output field.
3. Add an **Assignment** step to the process.
4. Click the **Assignments** tab.
5. In the **Field** column, select the output field you created.
6. In the **Assignment** column, enter the SQL query WHERE clause. For example, id = {$input.userid}
7. Save the assignment.

For example, to read a user with ID 78798, use the query id = 78798. The result is stored in the output field and can be used in subsequent process steps.

Write data to a Google BigQuery table

You can write data to a Google BigQuery table by inserting new records or updating existing records.

To insert new records, you use a **Create** step and map process fields to the table fields. The **Create** step writes the data to the table when the process runs.

To update existing records, you first query the row that you want to update and then use an **Assignment** step to modify specific field values. Use this approach to change one or more fields in a row without affecting other fields.

Note: You can perform update operations only on tables that have primary keys. Ensure that you have custom keys set up for the tables that you want to update.

Insert data into a Google BigQuery table

You can insert new records into a Google BigQuery table by using a **Create** step.

Perform the following sample steps to insert data into a Google BigQuery table:

1. In the Process Designer, open the process.
2. Add a **Create** step to the process.
3. In the **Create** step, click the **Object** field and select **Google BigQuery V2** connection.
4. In the **Connection** field, select the **Google BigQuery V2** connection.
5. In the **Type** field, select the object type that corresponds to your Google BigQuery table.
   For example, select **User** if you want to insert data into the **User** table.
6. Map process fields to the table fields.
   For example:
   - Map **{$input.userid}** to the **id** field
   - Map **userName** to the **name** field
   - Map **userAge** to the **age** field
7. Save the step.

Application Integration writes data to the Google BigQuery table when the process runs.

Example
To insert a new user with ID 456, name "John Smith", and age 30, map the following process fields to the table fields:

- The userId (value: 456) to the table field id
- The process field userName (value: "John Smith") to the table field name

- The process field userAge (value: 30) to the table field age

Update data in a Google BigQuery table

You can update existing data in a Google BigQuery table by first querying the row and then modifying the field values in an **Assignment** step.

Guidelines
Consider the following guidelines when you update a table using Google BigQuery V2 connections in a process:

- For update operations, tables must have primary keys defined.
- Map process fields to table fields accurately in **Create** steps.

Perform the following sample steps to update data in a Google BigQuery table:

1. Query the row that you want to update.
   In an **Assignment** step, use a SQL query to retrieve the row and assign it to a temporary process field.

   For example, to query a user with a specific ID, use id={$input.userid}.
2. Assign the query result to a process field such as $temp.user.
3. Add another **Assignment** step to update the field value.
4. In the **Field** column, select the field that you want to update by navigating through the temporary process field.
   For example, select $temp.user > name to update the user name.
5. In the **Assignment** column, enter the new value or map a process field that contains the new value.
   For example, enter newUserName to update the name field.
6. Save the assignment.

The table row is updated with the new value when the process executes.

Example
To update data in a Google BigQuery table, use the following example:

To update the name of a user with ID 123, you first query the user by using id = 123 and assign the result to $temp.user. Then, in an **Assignment** step, you select $temp.user > name and assign the new value "Jane Doe". The process updates the user's name to "Jane Doe" in the Google BigQuery table when it runs.

Delete data from a Google BigQuery table

You can delete records from a Google BigQuery table by first querying the row and then use a Delete step to remove it.

Perform the following sample steps to delete data from a Google BigQuery table:

1. Query the row that you want to delete.
2. In an **Assignment** step, use a SQL query to retrieve the row and assign it to a temporary process field.
   For example, use id={$input.userid} and assign the query result to a process field such as $temp.user.
3. Add a **Service** step.
   a. On the **Service** tab, select **System Service** as the service type. Select **Delete Object** as the action.
   b. On the **Input Fields** tab, select the connection that contains the object to delete. For example, Google BigQueryV2 connection.
      Similarly, enter the **Object Type** and **Object ID**.

      For example, select $temp.user.
4. Save the step.

Guidelines

Consider the following guidelines when you delete a table using Google BigQuery V2 connections in a process:

- For delete operations, tables must have primary keys defined.
- Always query the row before deleting to ensure you are deleting the correct record.
- Verify the row before you run the delete operation. The operation is permanent.

Example
To delete data in a Google BigQuery table, use the following example:

To delete a user with ID 123, you first query the user by using the clause id = 123 and assign the result to $temp.user. Then, in the **Service** step, select the Google BigQuery V2 connection and set the reference to $temp.user. The process deletes the user with ID 123 from the `User` table when it runs.

Google BigQuery V2 data types

Understand how Google BigQuery data types map to process field types when you use a Google BigQuery V2 connection in Application Integration.

The following table describes the Google BigQuery data types that you can use with Application Integration:

Google BigQuery data type	Description	Maps to process field type
ARRAY	Ordered list of values	Array or List
BIGNUMERIC	Exact numeric value with 76.76 digits of precision	Decimal or String
BOOL	Boolean value (true or false)	Boolean
BYTES	Variable-length binary data	Binary or String
DATE	Calendar date (year, month, day)	Date
DATETIME	Date and time	DateTime
FLOAT64	64-bit floating-point number	Double or Float
GEOGRAPHY	Geographic data (points, lines, polygons)	String in well-known text (WKT) format
INT64	64-bit signed integer	Integer or Long
NUMERIC	Exact numeric value with 38 digits of precision	Decimal or String
STRING	Variable-length character data	String
STRUCT	Container of ordered fields	Object or Struct
TIME	Time of day, independent of date	String or Time

Google BigQuery data type	Description	Maps to process field type
TIMESTAMP	Absolute point in time with microsecond precision	DateTime or Timestamp

Consider the following factors when you work with Google BigQuery data types:

- **Timestamp precision**: Google BigQuery TIMESTAMP values have microsecond precision. When you map to DateTime fields, the precision might reduce to milliseconds.
- **Numeric precision**: For NUMERIC values that require precision, use String fields to avoid rounding errors.
- **Complex types**: For ARRAY and STRUCT types, ensure your process fields are configured as complex objects that match the structure.
- **DATE and TIME**: Google BigQuery separates date and time types. Ensure you use the appropriate field type in your process.
- **NULL values**: All Google BigQuery data types accept NULL values. Ensure your process handles null values appropriately.

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